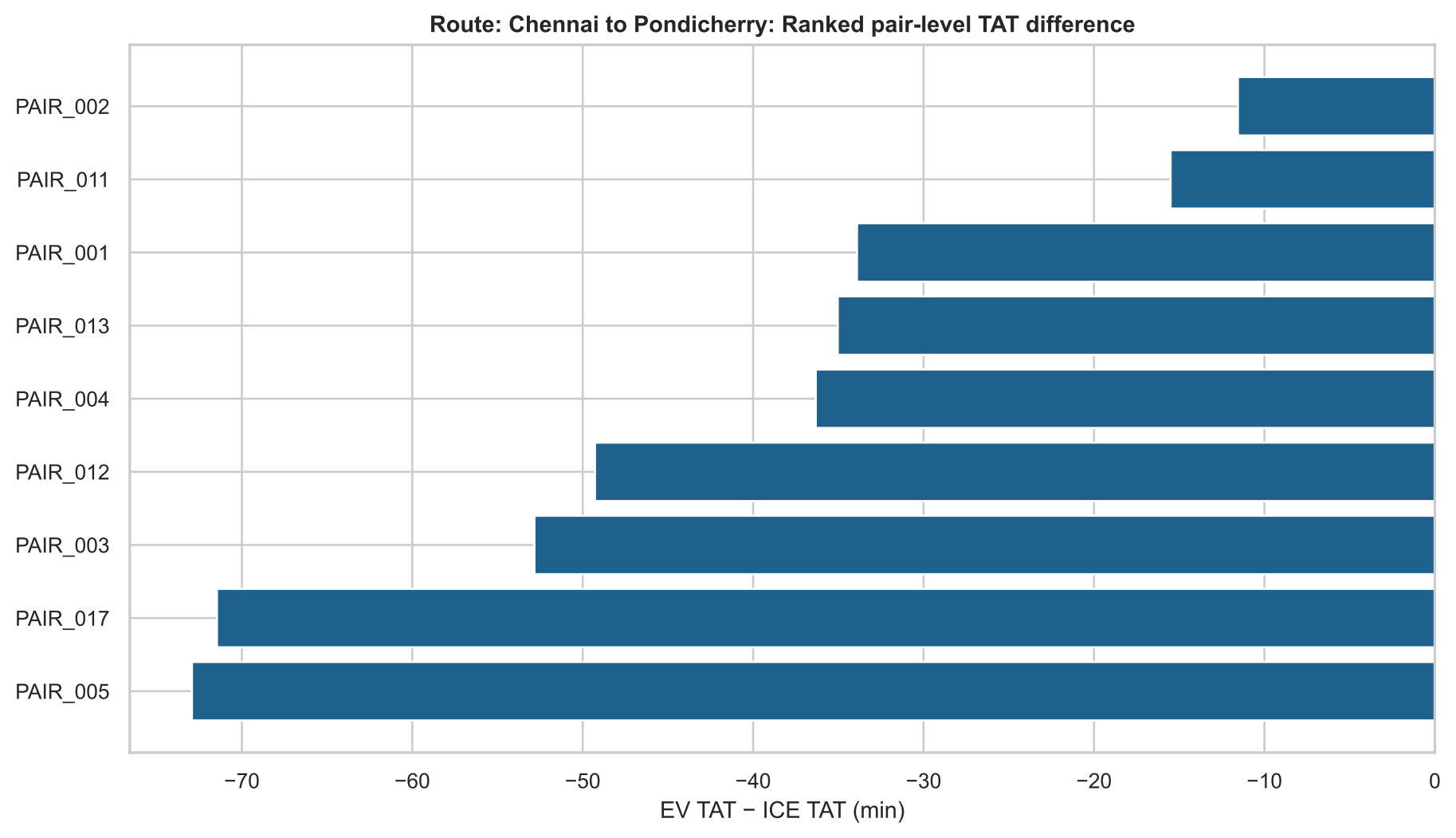
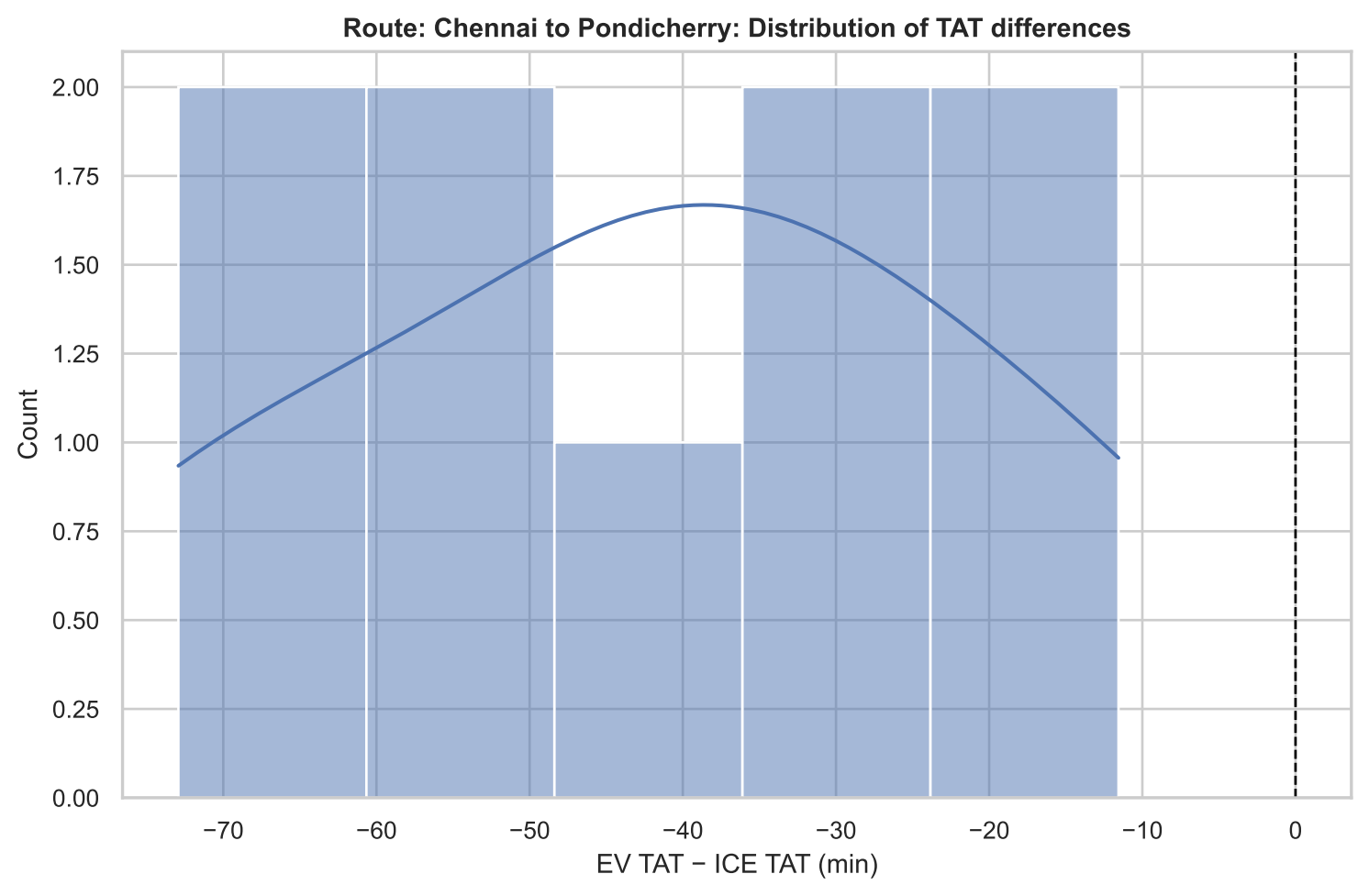
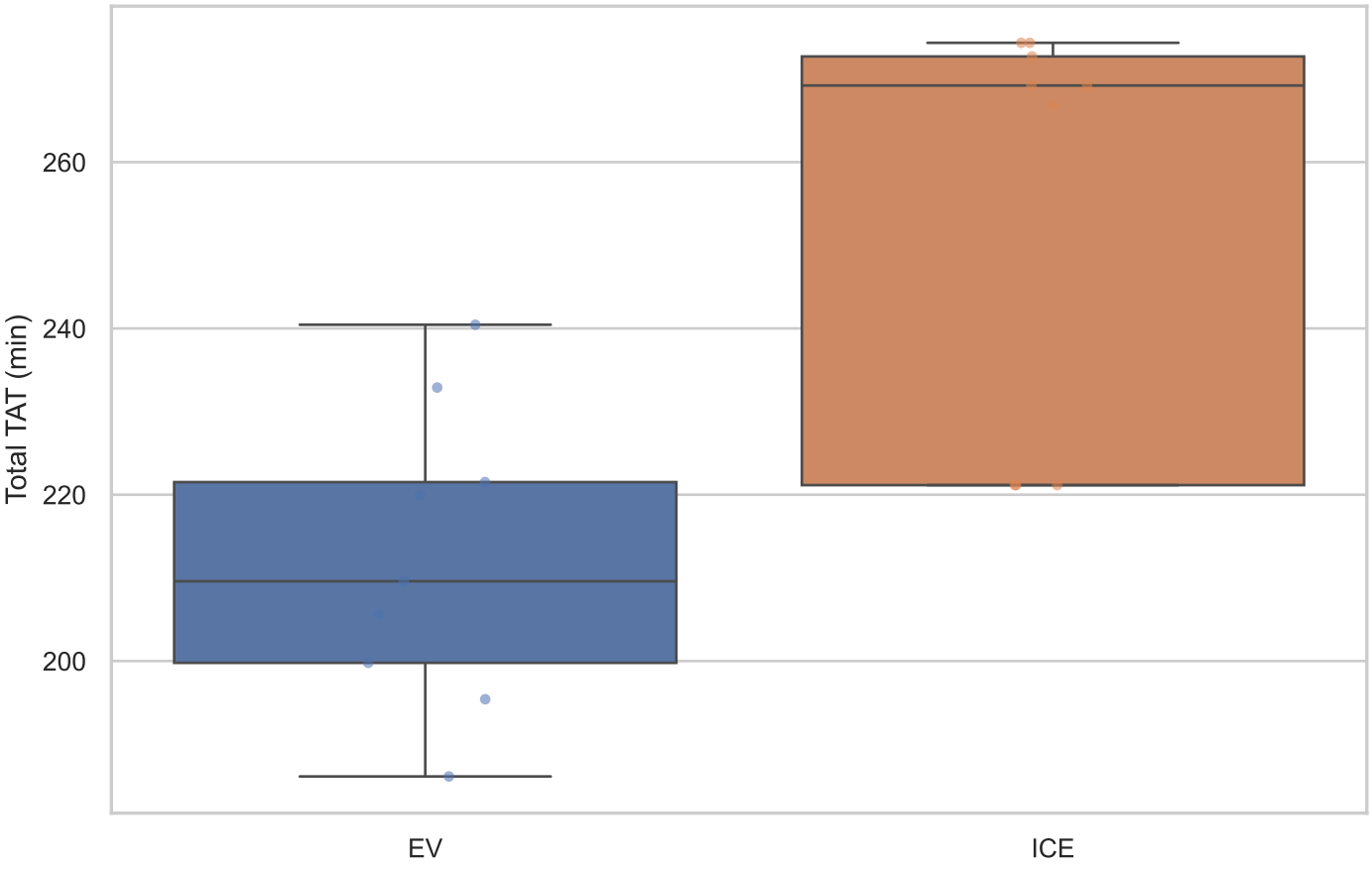
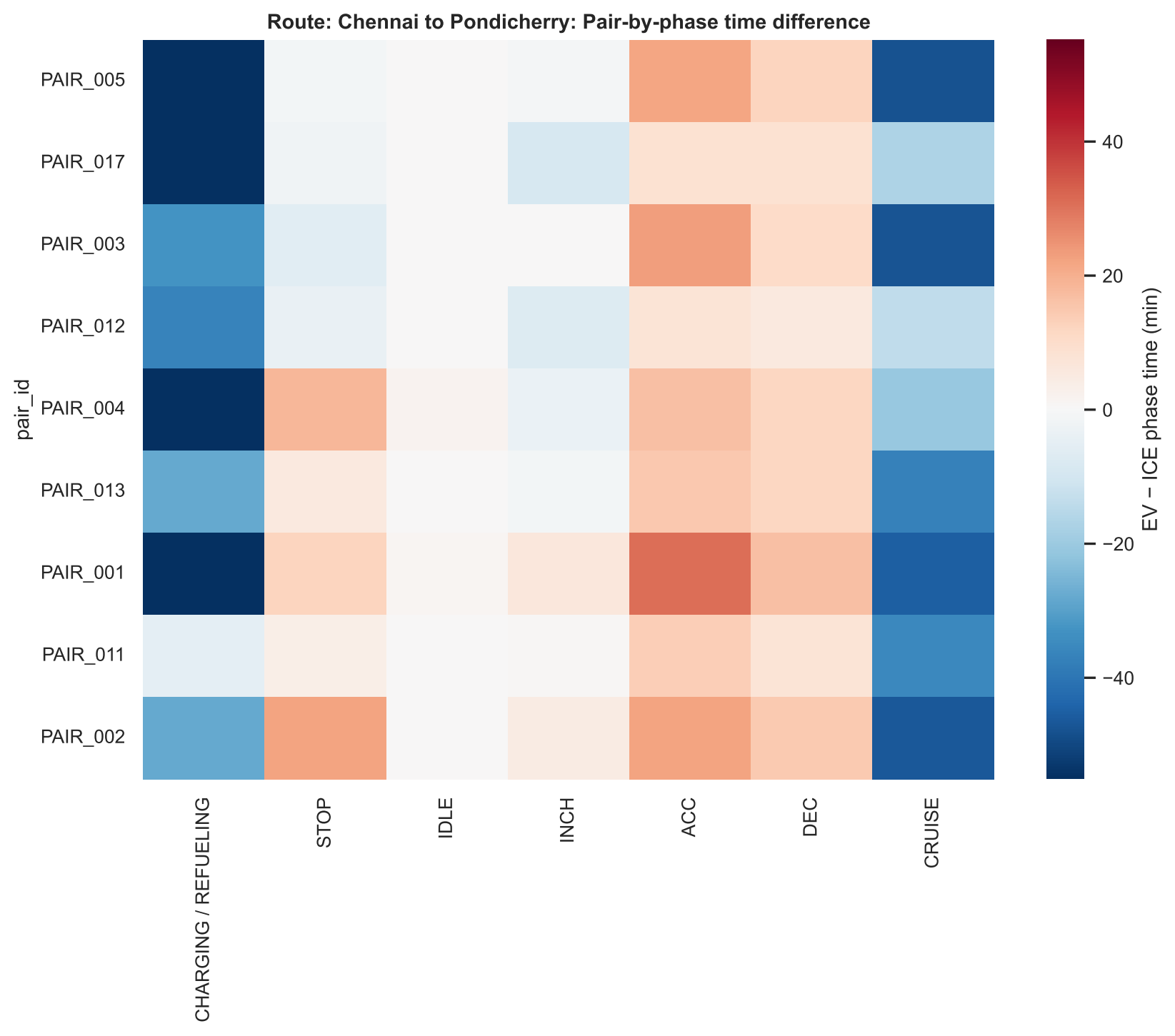
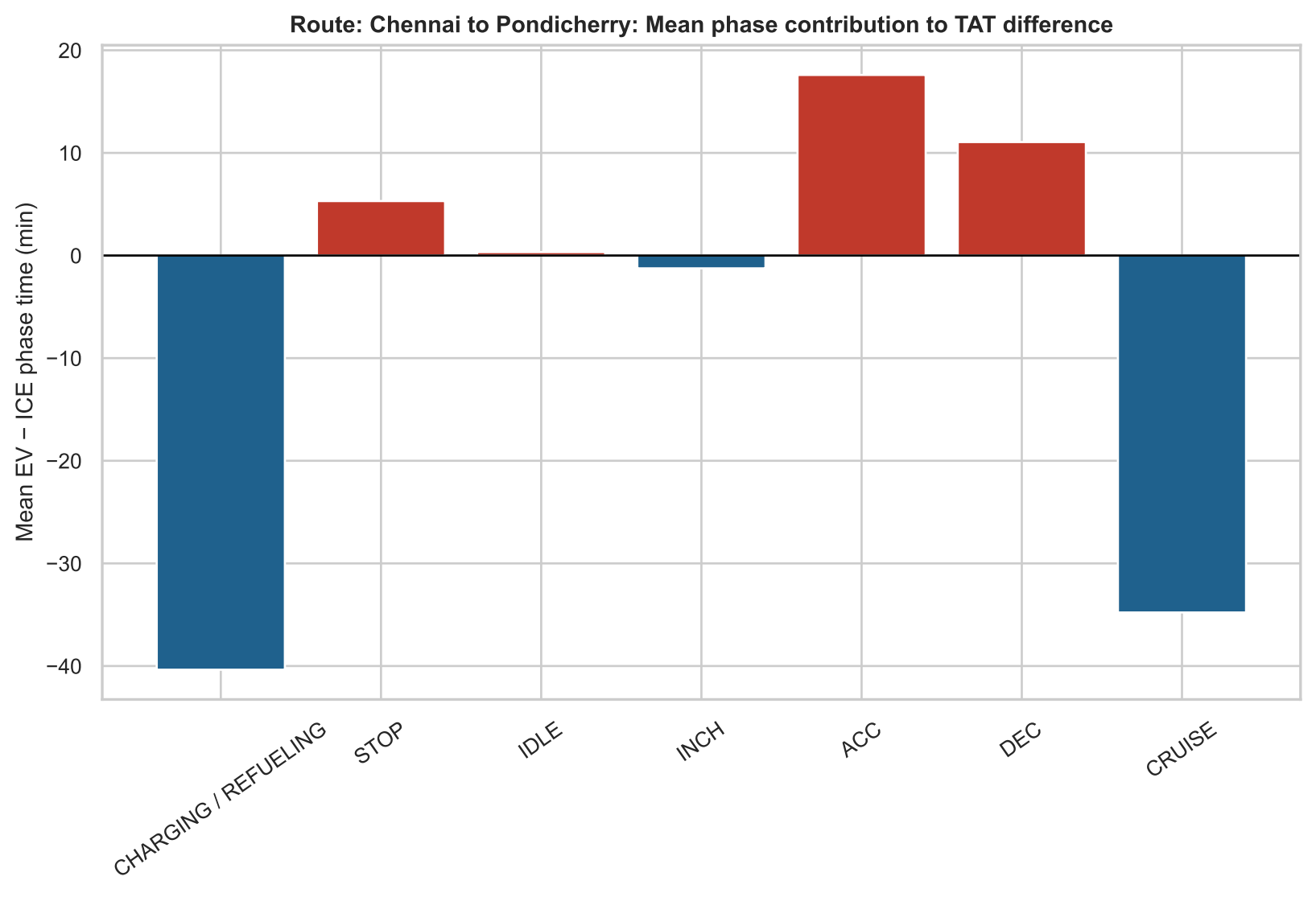
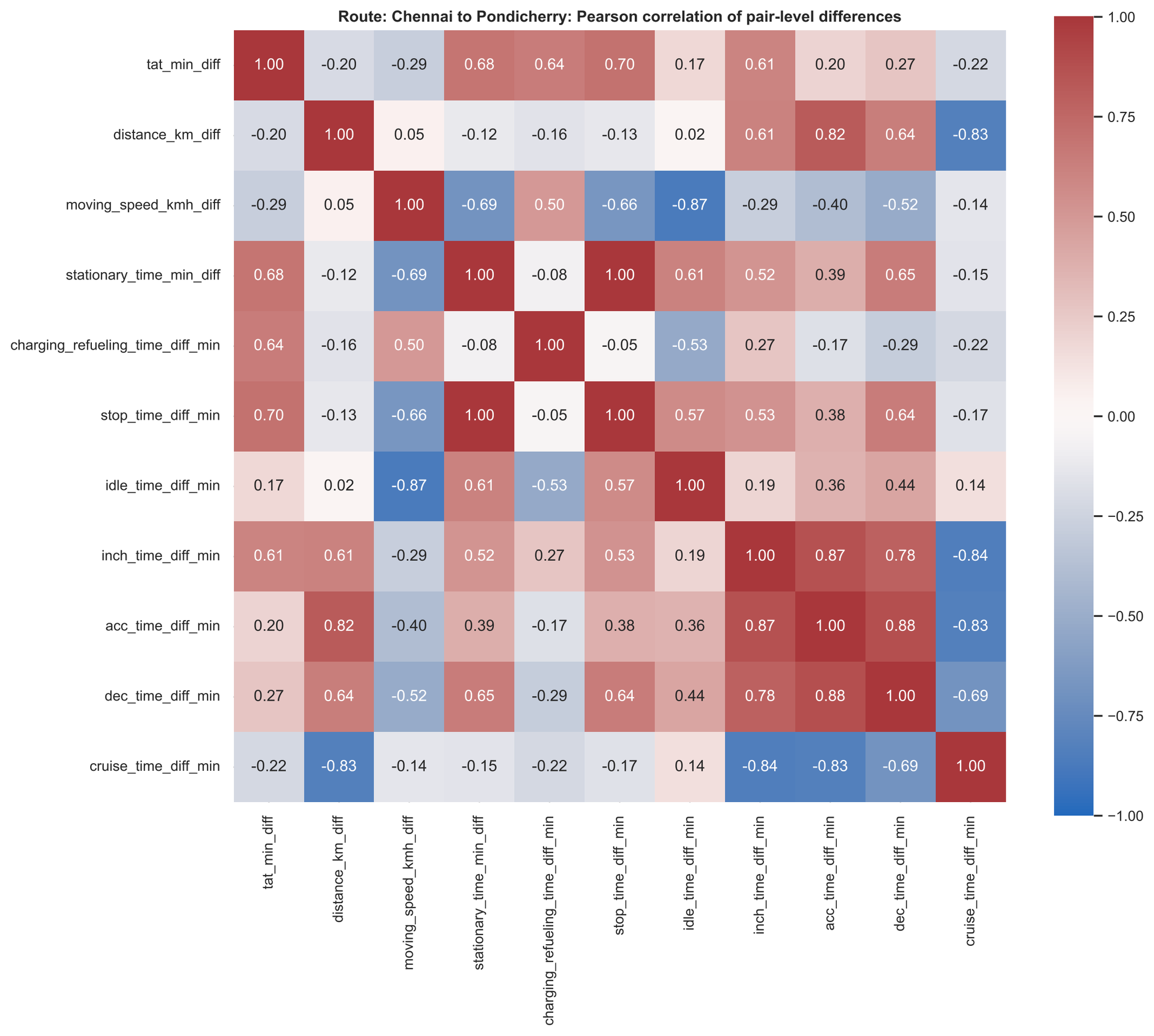

Route: Chennai to Pondicherry: Ranked pair-level TAT difference
PAIR_002
PAIR_011
PAIR_001
PAIR_013
PAIR_004
PAIR_012
PAIR_003
PAIR_017
PAIR_005
−70
−60
−50
−40
−30
−20
−10
0
EV TAT − ICE TAT (min)

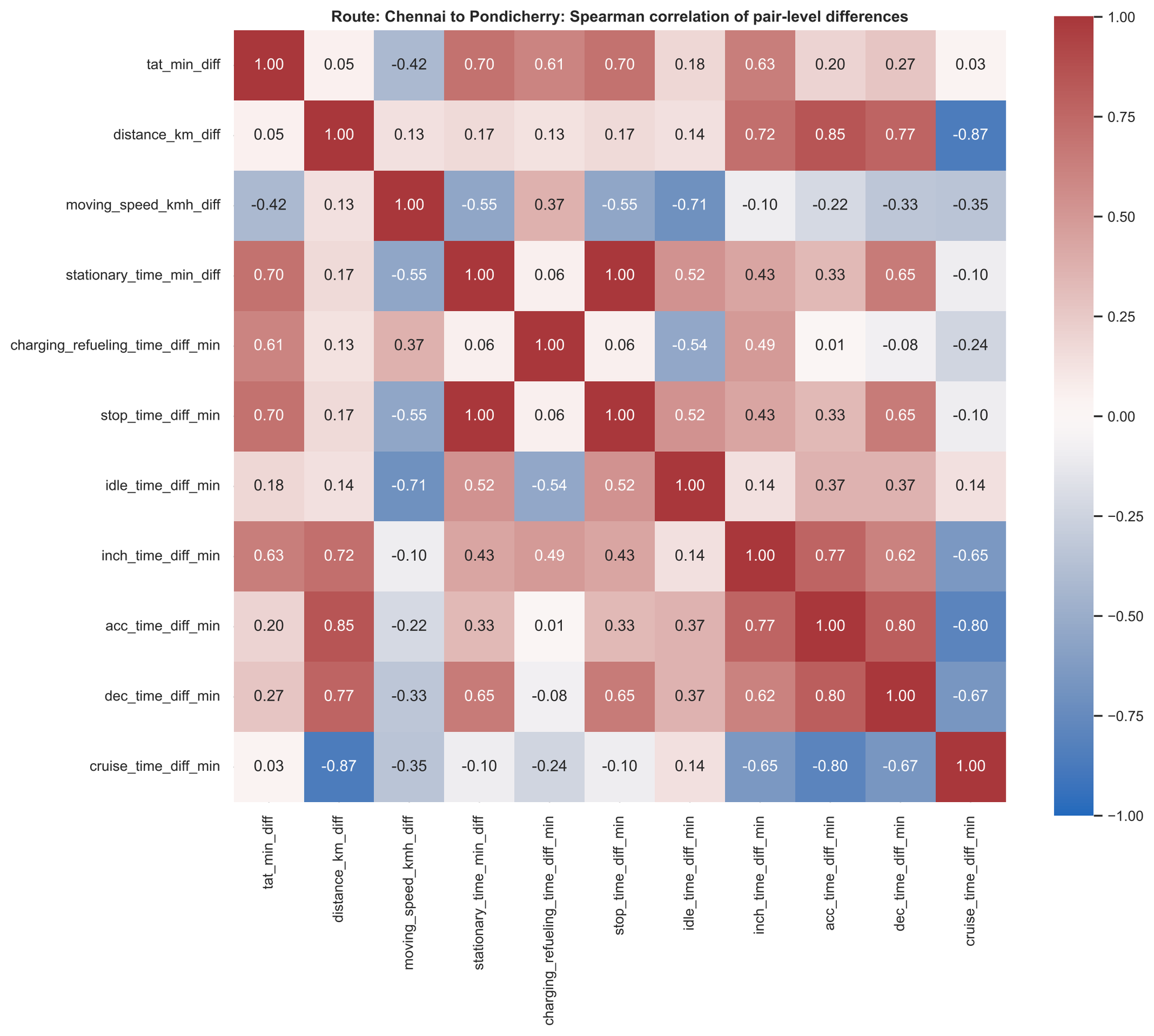

Route: Chennai to Pondicherry: Distribution of TAT differences
Count
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−70
−60
−50
−40
−30
−20
−10
0
EV TAT − ICE TAT (min)

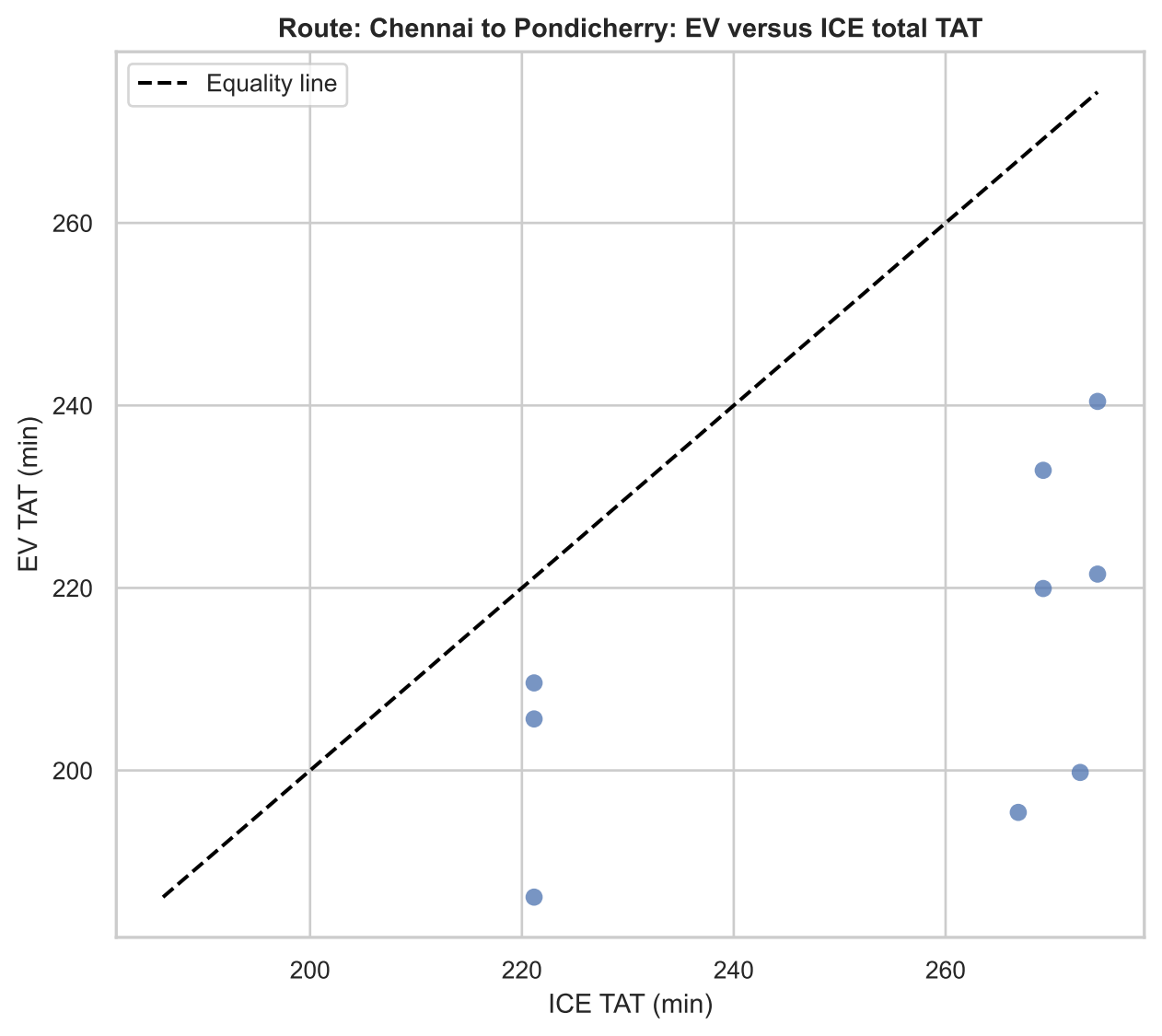

Route: Chennai to Pondicherry: EV and ICE TAT distribution
Total TAT (min)
260
240
220
200
EV
ICE

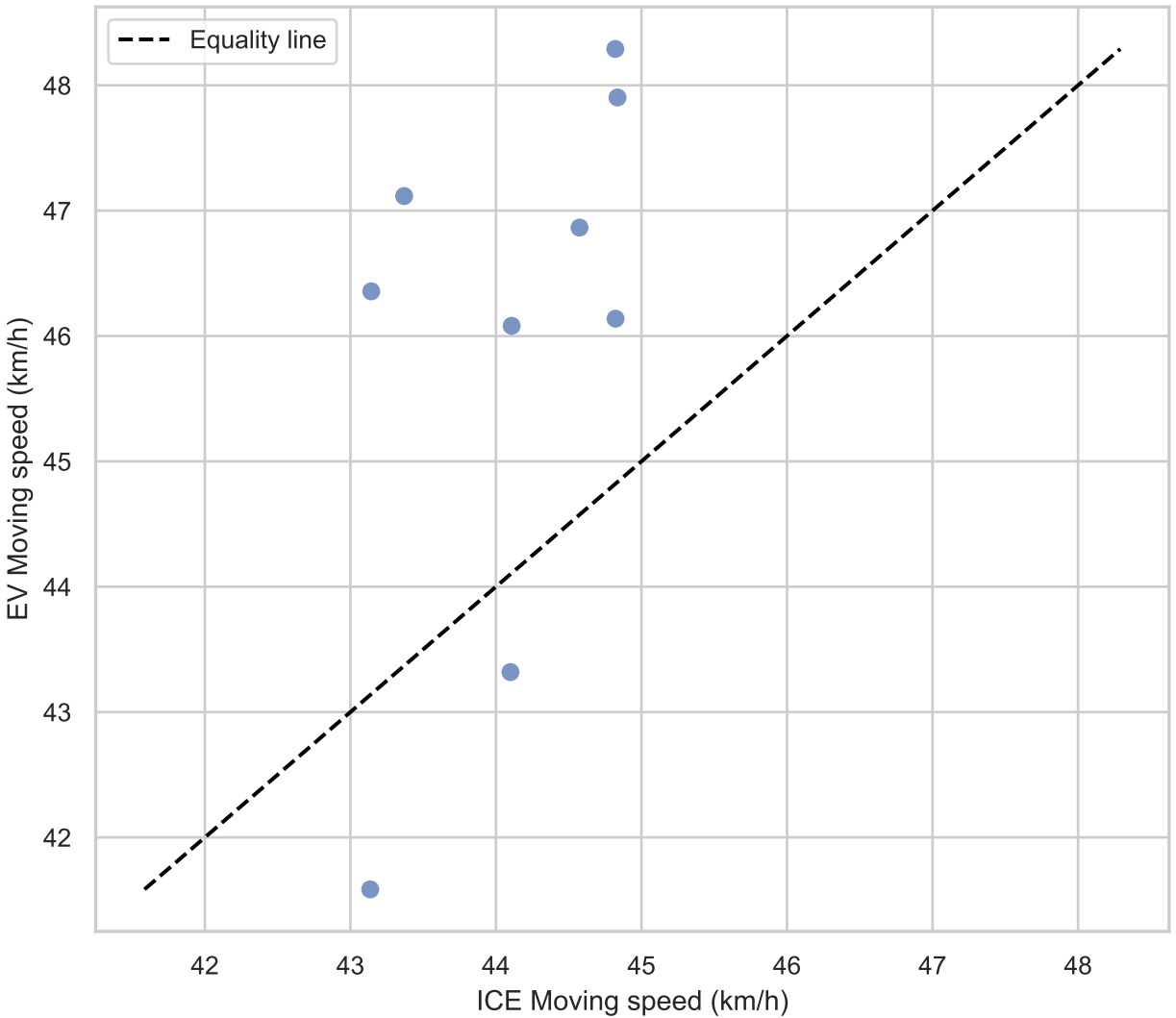

Route: Chennai to Pondicherry: Pair-by-phase time difference
pair_id
PAIR_005
PAIR_017
PAIR_003
PAIR_012
PAIR_004
PAIR_013
PAIR_001
PAIR_011
PAIR_002
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE
EV − ICE phase time (min)
40
20
0
−20
−40

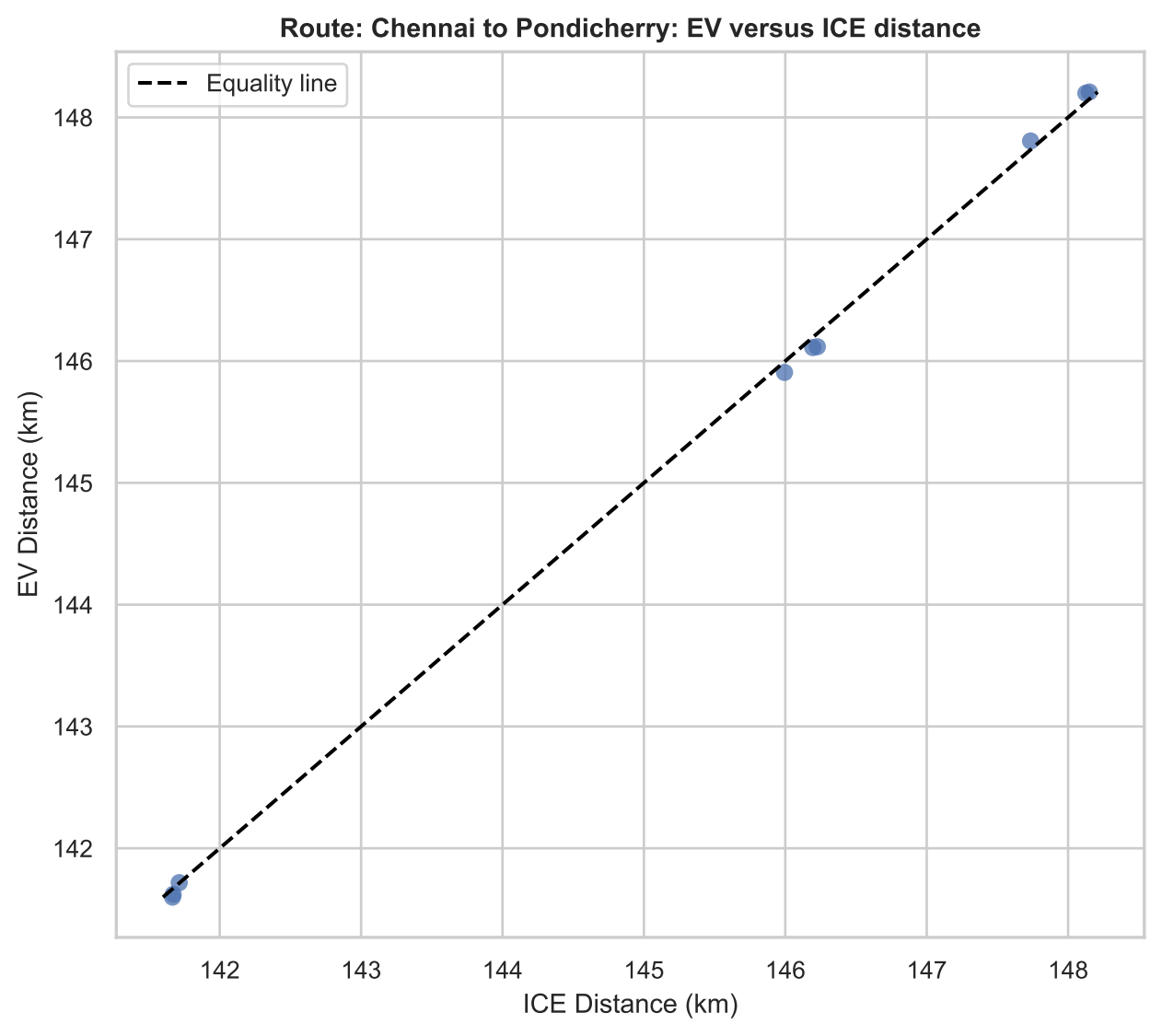

Route: Chennai to Pondicherry: Mean phase contribution to TAT difference
Mean EV – ICE phase time (min)
20
10
0
−10
−20
−30
−40
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE

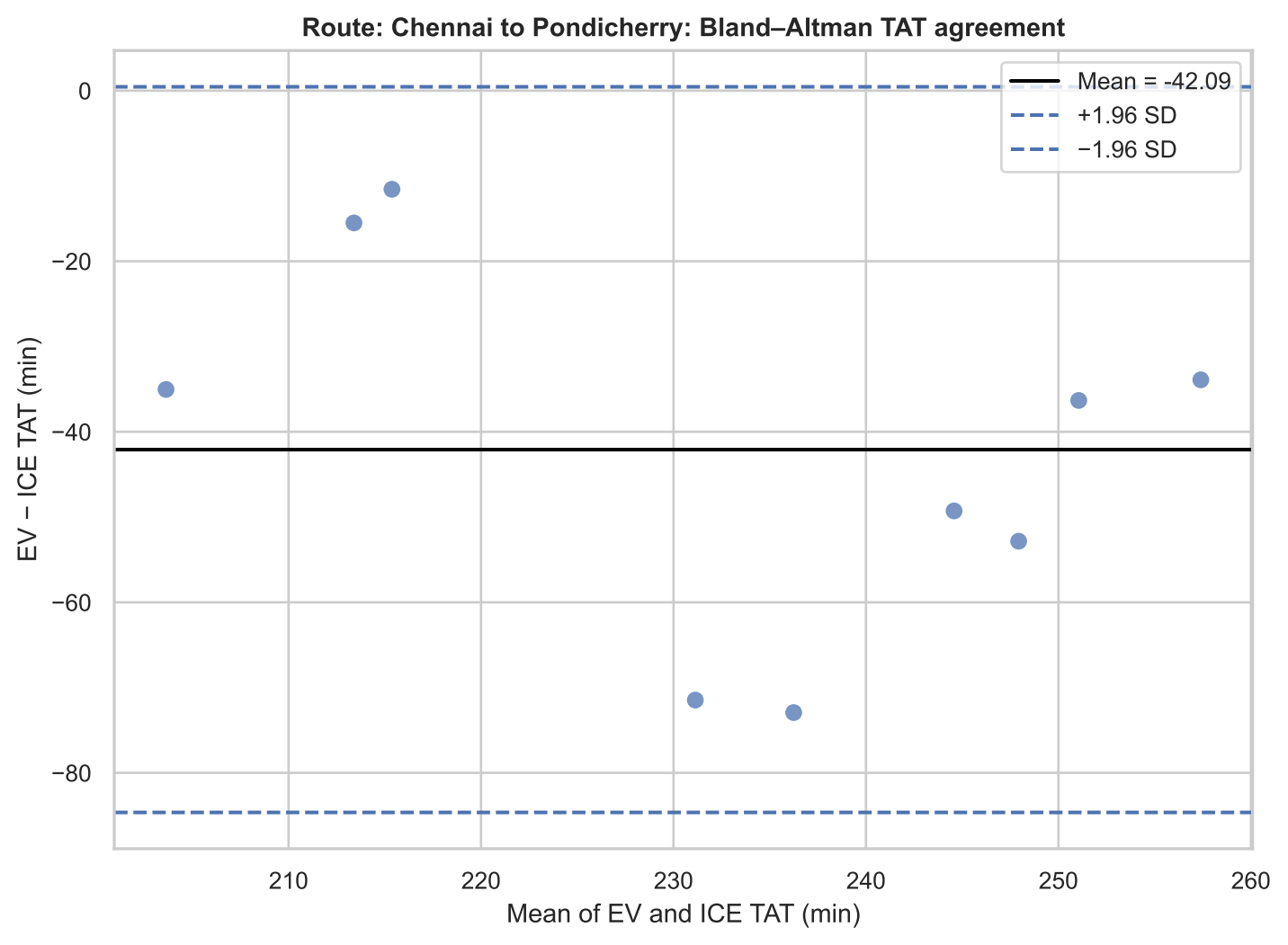

Route: Chennai to Pondicherry: Pearson correlation of pair-level differences
tat_min_diff 1.00 -0.20 -0.29 0.68 0.64 0.70 0.17 0.61 0.20 0.27 -0.22
distance_km_diff -0.20 1.00 0.05 -0.12 -0.16 -0.13 0.02 0.61 0.82 0.64 -0.83
moving_speed_kmh_diff -0.29 0.05 1.00 -0.69 0.50 -0.66 -0.87 -0.29 -0.40 -0.52 -0.14
stationary_time_min_diff 0.68 -0.12 -0.69 1.00 -0.08 1.00 0.61 0.52 0.39 0.65 -0.15
charging_refueling_time_diff_min 0.64 -0.16 0.50 -0.08 1.00 -0.05 -0.53 0.27 -0.17 -0.29 -0.22
stop_time_diff_min 0.70 -0.13 -0.66 1.00 -0.05 1.00 0.57 0.53 0.38 0.64 -0.17
idle_time_diff_min 0.17 0.02 -0.87 0.61 -0.53 0.57 1.00 0.19 0.36 0.44 0.14
inch_time_diff_min 0.61 0.61 -0.29 0.52 0.27 0.53 0.19 1.00 0.87 0.78 -0.84
acc_time_diff_min 0.20 0.82 -0.40 0.39 -0.17 0.38 0.36 0.87 1.00 0.88 -0.83
dec_time_diff_min 0.27 0.64 -0.52 0.65 -0.29 0.64 0.44 0.78 0.88 1.00 -0.69
cruise_time_diff_min -0.22 -0.83 -0.14 -0.15 -0.22 -0.17 0.14 -0.84 -0.83 -0.69 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

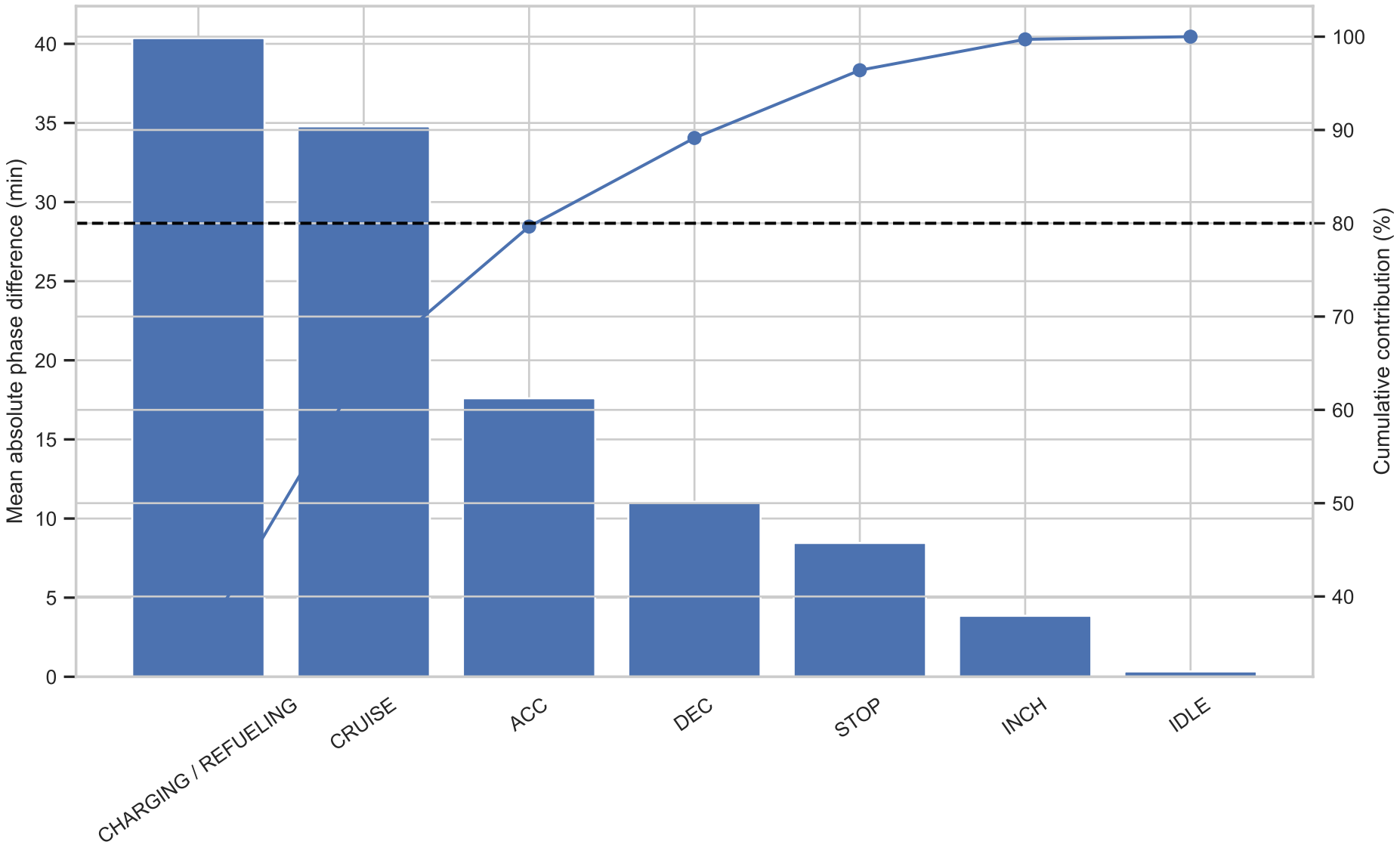

Route: Chennai to Pondicherry: Spearman correlation of pair-level differences
tat_min_diff 1.00 0.05 -0.42 0.70 0.61 0.70 0.18 0.63 0.20 0.27 0.03
distance_km_diff 0.05 1.00 0.13 0.17 0.13 0.17 0.14 0.72 0.85 0.77 -0.87
moving_speed_kmh_diff -0.42 0.13 1.00 -0.55 0.37 -0.55 -0.71 -0.10 -0.22 -0.33 -0.35
stationary_time_min_diff 0.70 0.17 -0.55 1.00 0.06 1.00 0.52 0.43 0.33 0.65 -0.10
charging_refueling_time_diff_min 0.61 0.13 0.37 0.06 1.00 0.06 -0.54 0.49 0.01 -0.08 -0.24
stop_time_diff_min 0.70 0.17 -0.55 1.00 0.06 1.00 0.52 0.43 0.33 0.65 -0.10
idle_time_diff_min 0.18 0.14 -0.71 0.52 -0.54 0.52 1.00 0.14 0.37 0.37 0.14
inch_time_diff_min 0.63 0.72 -0.10 0.43 0.49 0.43 0.14 1.00 0.77 0.62 -0.65
acc_time_diff_min 0.20 0.85 -0.22 0.33 0.01 0.33 0.37 0.77 1.00 0.80 -0.80
dec_time_diff_min 0.27 0.77 -0.33 0.65 -0.08 0.65 0.37 0.62 0.80 1.00 -0.67
cruise_time_diff_min 0.03 -0.87 -0.35 -0.10 -0.24 -0.10 0.14 -0.65 -0.80 -0.67 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

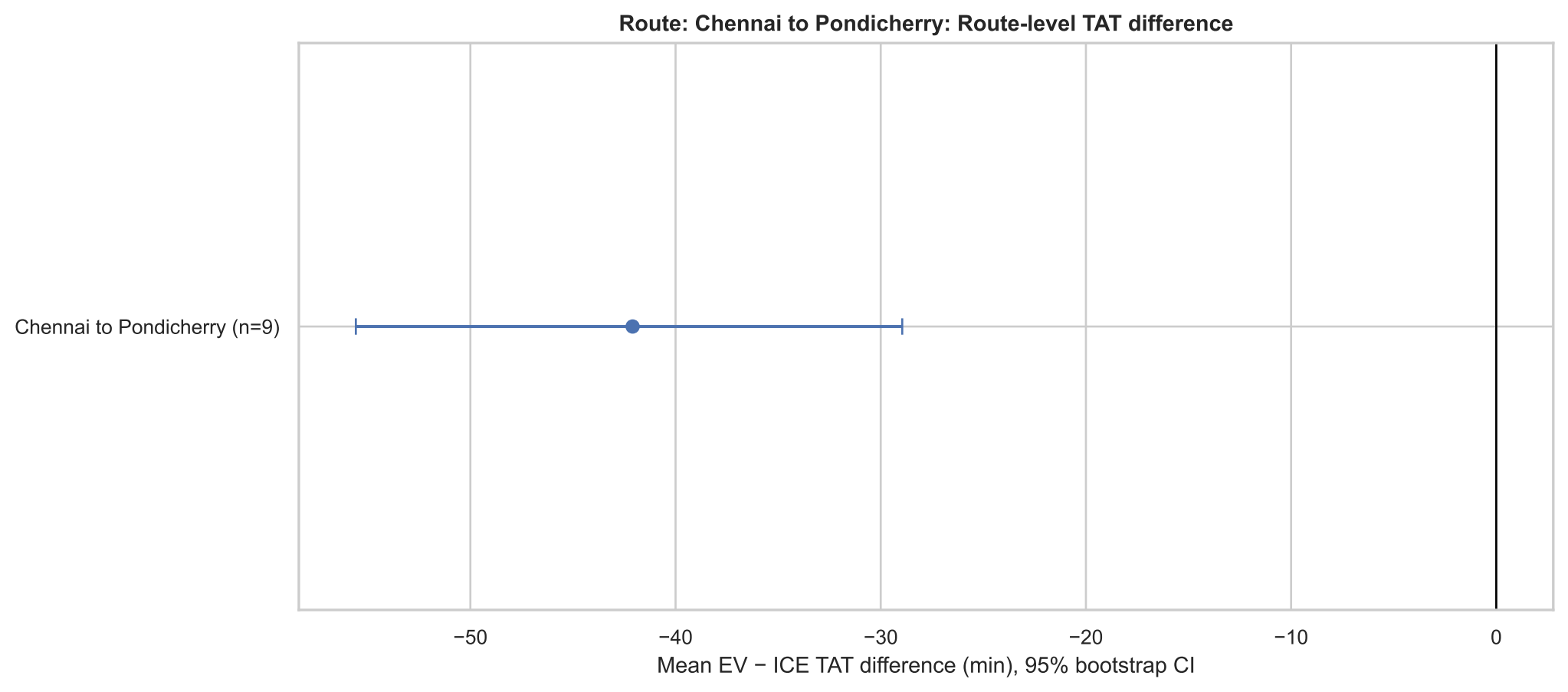

Route: Chennai to Pondicherry: EV versus ICE total TAT
Equality line
EV TAT (min)
ICE TAT (min)
200
220
240
260
200
220
240
260

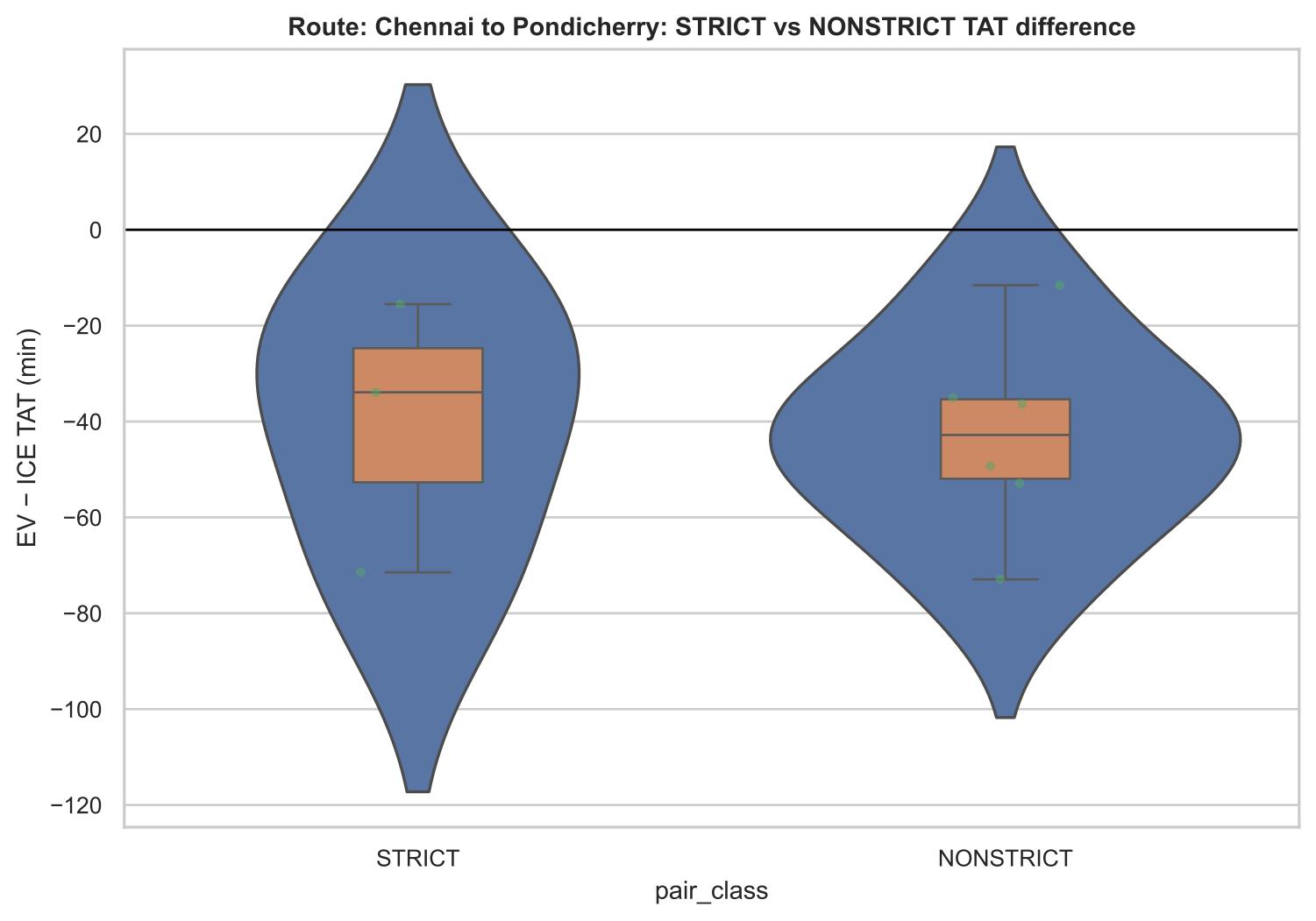

Route: Chennai to Pondicherry: EV versus ICE moving speed
Equality line
EV Moving speed (km/h)
ICE Moving speed (km/h)
42
43
44
45
46
47
48

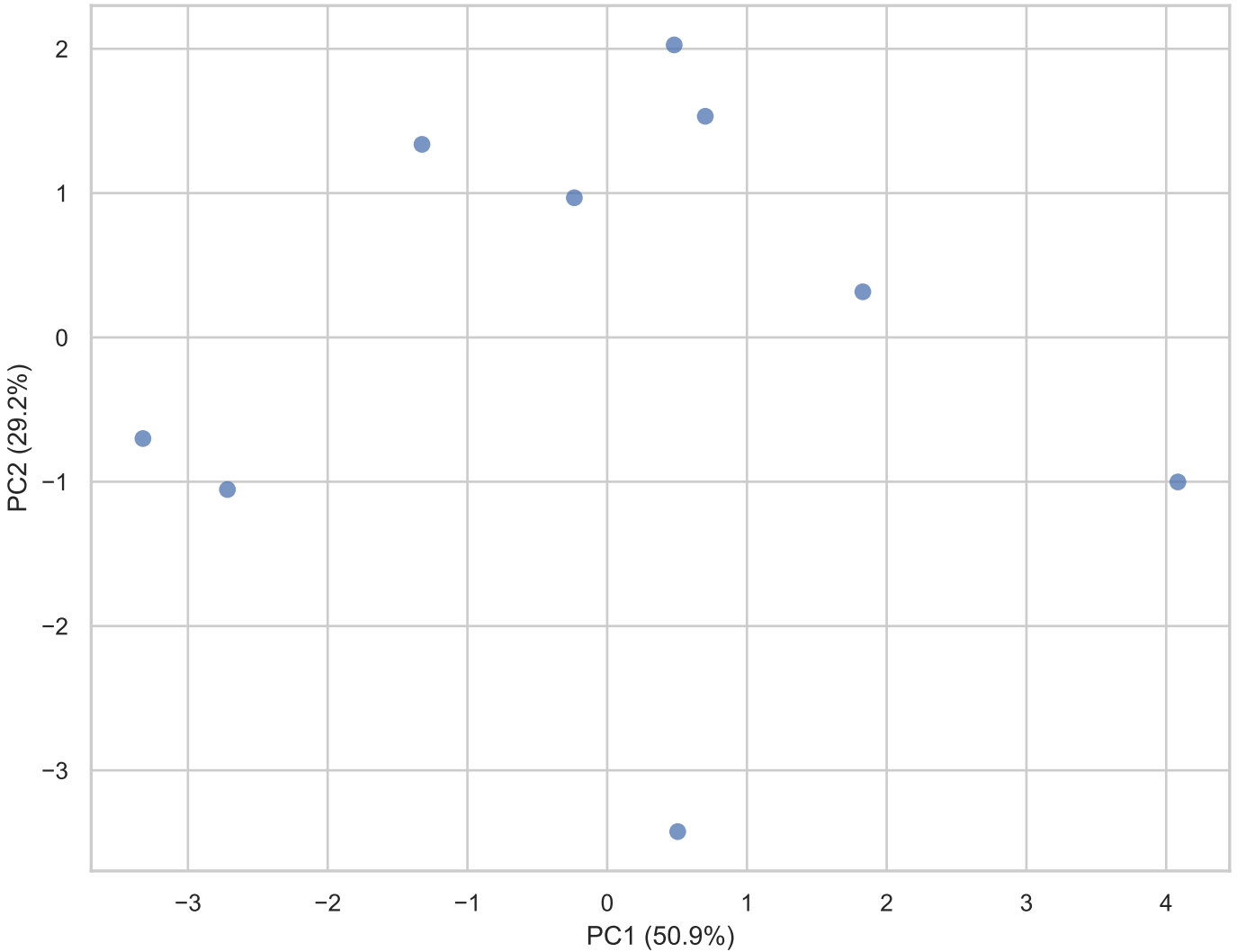

Route: Chennai to Pondicherry: EV versus ICE distance
Equality line
EV Distance (km)
ICE Distance (km)
142
143
144
145
146
147
148

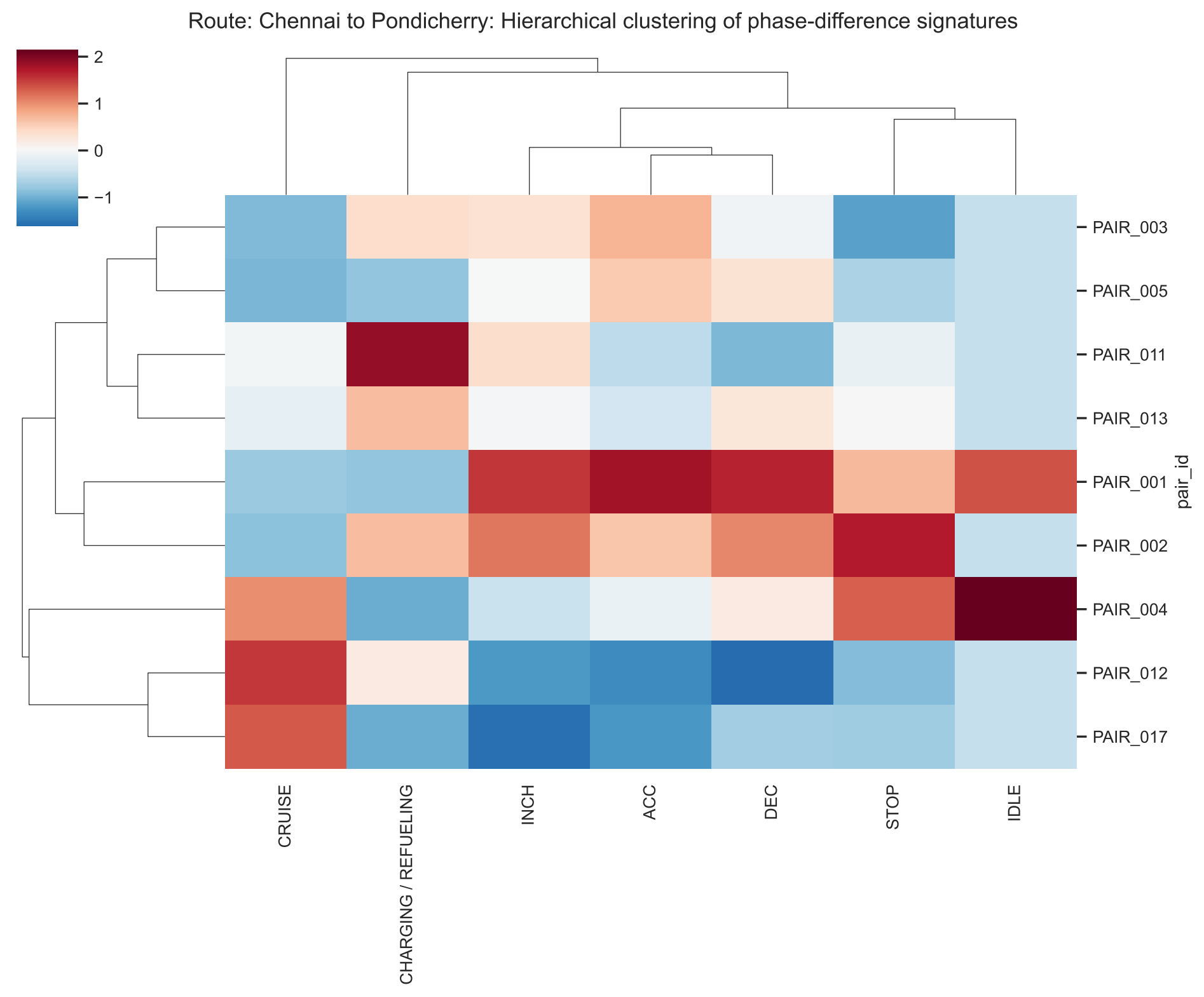

Route: Chennai to Pondicherry: Bland–Altman TAT agreement
Mean = -42.09
+1.96 SD
−1.96 SD
0
−20
−40
−60
−80
EV – ICE TAT (min)
210
220
230
240
250
260
Mean of EV and ICE TAT (min)


Route: Chennai to Pondicherry: Pareto analysis of phase effects
Mean absolute phase difference (min)
Cumulative contribution (%)
0
5
10
15
20
25
30
35
40
40
50
60
70
80
90
100
CHARGING / REFUELING
CRUISE
ACC
DEC
STOP
INCH
IDLE


Route: Chennai to Pondicherry: Route-level TAT difference
Chennai to Pondicherry (n=9)
−50
−40
−30
−20
−10
0
Mean EV − ICE TAT difference (min), 95% bootstrap CI


Route: Chennai to Pondicherry: STRICT vs NONSTRICT TAT difference
20
0
−20
−40
−60
−80
−100
−120
EV – ICE TAT (min)
STRICT
NONSTRICT
pair_class


Route: Chennai to Pondicherry: PCA of EV−ICE operational differences
PC2 (29.2%)
PC1 (50.9%)
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3
4


Route: Chennai to Pondicherry: Hierarchical clustering of phase-difference signatures
2
1
0
−1
PAIR_003
PAIR_005
PAIR_011
PAIR_013
PAIR_001
PAIR_002
PAIR_004
PAIR_012
PAIR_017
pair_id
CRUISE
CHARGING / REFUELING
INCH
ACC
DEC
STOP
IDLE